# Righteousness Team Design Document
## Robocup 2024 Worldwide

## Junior League - Lightweight Soccer

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## Abstract

Our team designed and built advanced robots for the RoboCup Junior lightweight soccer competition, achieving significant improvements in functionality and efficiency. This design document provides detailed insights into our innovative approach, such as using robust sensors and precise control systems. Our robots are equipped with state-of-the-art hardware. We also implemented an effective strategy by distinctly separating the roles of an attacking robot (the striker) and a defending robot (the goalkeeper). By utilizing 3D printing technology, we created various components, which resulted in weight reduction. This design document overturns common perceptions in robot design by demonstrating how strategic hardware and software integration can lead to superior performance in robotic competitions.

**Table of content:**

## Introduction

We are the team Righteousness, which is a dedicated team based in Macau China. Our team members are studying at Macau Pui Ching Middle School, which has been actively participating in RoboCup for three years. Last year, we had the fortunate opportunity to travel to Japan, where we gained invaluable experience.

Team photo

Roles of Team Members

| Thomas | Zachary | Marco | Jayden |
|---|---|---|---|
| Circuit | Mechanic | Program | Program |
| Design | Design | 3D Printer | Communication |

While it is beneficial for every team member to understand their specific roles in each process, our learning approach requires us to gain knowledge across all areas. Therefore, this section highlights the primary responsibilities of each team member. Followed by budget considerations, they are crucial since we do not have ample funds to create everything in advance. Therefore, we compiled a table listing all necessary materials. The chart below outlines our bill of materials, detailing each item and its associated costs.

Bill of materials

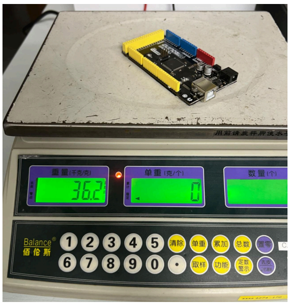
| Material | | quantity | Price for each(USD) | Price |
|---|---|---|---|---|
| Structural | fiberglass | 8 | 6.89 | 55.09 |
| | PolyLite™ PLA | 1 | 16.53 | 16.53 |
| | PolyLite™ ASA | 1 | 14.79 | 14.79 |
| | Polyoxymethylene | 1 | 13.77 | 13.77 |
| | PCB | 1 | 21.74 | 21.74 |
| Power system | 12V Lithium Battery | 10 | 15.84 | 158.4 |
| | Battery Charger | 1 | 134.97 | 134.97 |
| | Balanced charging plugs | 10 | 3.31 | 33.1 |
| | Wire | 1 | 10 | 10 |
| Hardware(for 2 robot) | | | | |
| engines system | dc motor | 8 | 137.72 | 1101.76 |
| | Dribbling dc motor | 2 | 27.54 | 55.08 |
| | Kick | 2 | 6.89 | 13.78 |
| | H-bridge | 4 | 12.39 | 49.56 |
| Sensoring | Compass | 2 | 123.95 | 247.9 |
| | IR seeker | 4 | 165.25 | 661 |
| | Greyscale | 8 | 8.26 | 66.08 |
| | ultrasonic | 8 | 12.39 | 99.12 |
| Electrical | Arduino Mega 2560 pro | 2 | 8.26 | 16.52 |
| | nano | 2 | 5.51 | 11.02 |
| | | | Total Price | 2780.21 |

# Development

## Hardware - Sensors and Components:

1. Main controller - Mega 2560 Pro

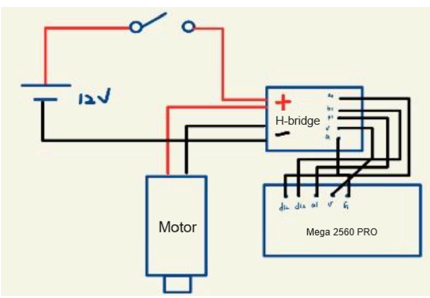
Last year, our team used the Arduino Mega 2560, but after the construction of various sensors and devices, the robots were overweight. Therefore, we started to decrease the weight in different aspects. As a result, the Mega Pro is an embedded controller with a smaller size (38mm x 55mm) than Arduino Mega 2560 (103mmx55mm). With the same number of ports as Arduino Mega 2560, yet 63% smaller in size and 73.5% in weight, it was chosen as the main controller.

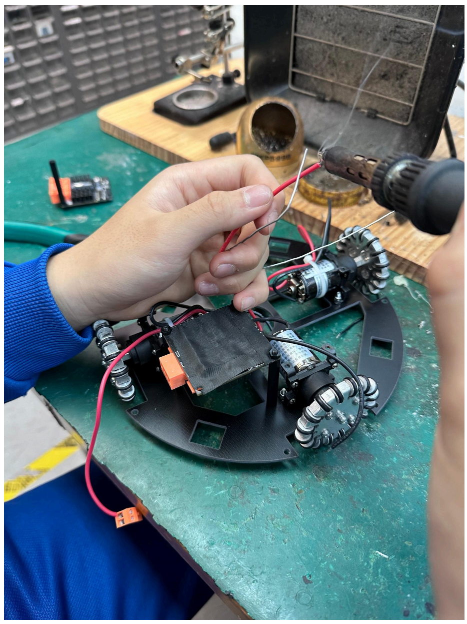




Arduino Mega 2560 pro Arduino Mega 2560

2. Engine system
   a) H-bridge

When selecting motors, speed is a crucial factor to consider. Ultimately, we opted for 12V DC motors due to their performance capabilities. However, the robot's PCB system cannot handle the excessive electrical volume directly. To address this, we integrated an H-bridge. The following figure illustrates the engine system circuit: a 12V battery is connected to both the H-bridge and the motor, while the Mega board is connected to the H-bridge via PWM, digital pins, 5V, and ground.



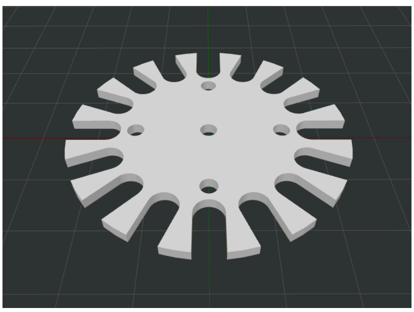
   b) DC motor

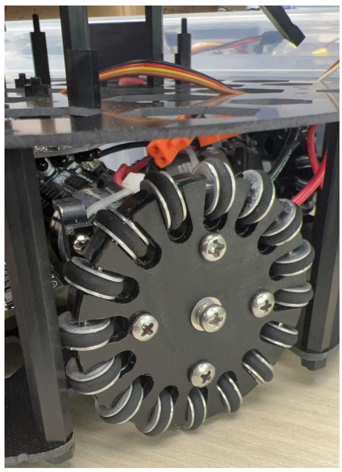
For our motors, we selected the JMP-BE-3561, a 12V DC motor with a power consumption of 8.4W and a rotation speed of approximately 1700 rpm. This motor is noted for its high power and density efficiency, making it particularly suitable for high-performance robot football systems. The JMP-BE-3561 is a popular choice in RoboCupJunior soccer games, offering a robust driving force without excessive battery consumption.

However, this motor does have its drawbacks. Each motor weighs 83.8g, and using four of them significantly impacts the total weight, accounting for roughly 25% of the overall mass. While the motors provide substantial power, their weight necessitates careful consideration in the robot's overall design and balance.


c) Omni-wheels

To enable the robot to move quickly and efficiently around the field, we use Omni-wheels, which allow for omnidirectional movement by adjusting the speed and direction of each motor. Additionally, we selected POM boards for their hardness and lightweight properties. The design of the wheels was meticulously crafted using CorelDRAW to ensure optimal performance and durability.





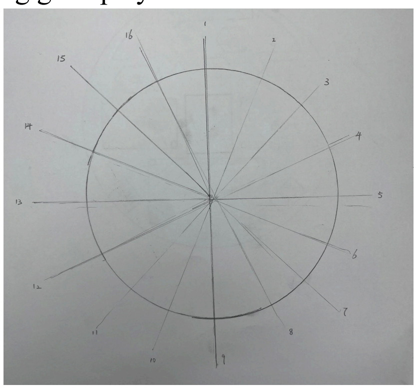
3. Sensors

a) Compass

To ensure the robot correctly faces the goal while scoring, preventing it from kicking the ball into our own goal, we incorporated a compass to calculate its orientation. A BE-2617 compass is mounted on the top plate and connected via the I2C protocol. This direction sensor, specifically designed for robots, determines the robot's facing direction by measuring the surrounding magnetic field. The compass provides directional values from 0 to 360 degrees, allowing us to set new directions for various practical applications and enhance reading precision through hardware compensation. The status indicator, consisting of a circle of LEDs, provides a clear visual of the compass's operational status, making it easy to monitor.

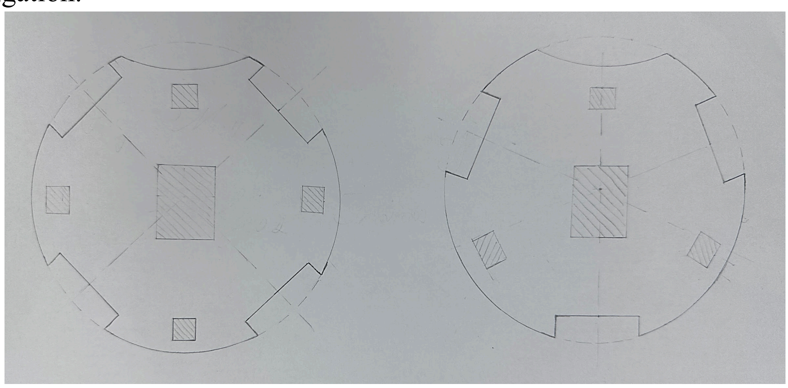
b) IR seeker

To identify the ball, we use two IR sensors: the locator and the seeker, both connected via the I2C protocol. The locator provides information about the ball's position relative to the robot, represented on a 360-degree grid. Based on the distance to the ball, the robot can adjust its path; the closer the ball, the greater the curvature of the motion. This system allows the robot to effectively maneuver around the ball, ensuring precise control during gameplay..



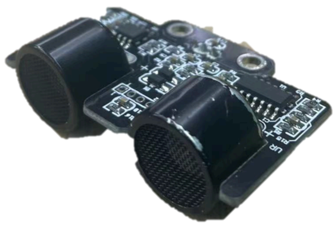
c) Grayscale

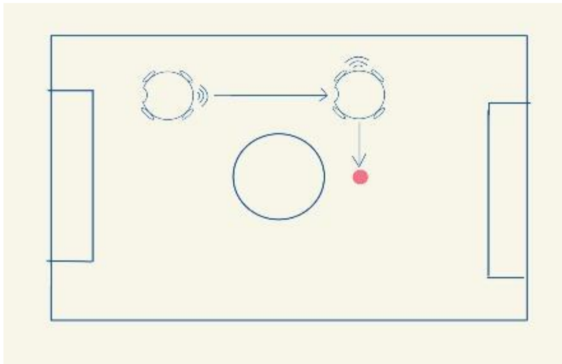
Initially, we considered using three grayscale sensors, but found it challenging to accurately determine the robot's surroundings. Due to this limitation and motor constraints, we decided to place four grayscale sensors on our bottom plate to detect the lines more effectively. These sensors consist of photosensitive transistors and light-emitting diodes, with data collected through analog ports. To guide the robot away from the lines, we applied the cosine rule, utilizing mathematical vector plotting for precise navigation.



d) Ultrasonic

We use ultrasonic sensors to aid in both locating and attacking the goal. These sensors help the robot navigate to specific locations by measuring the distance between the ultrasonic sensor and the walls, enabling efficient movement around the field. During the attacking phase, the robot can adjust its angle to continuously face the goal, ensuring precise targeting and alignment for scoring.

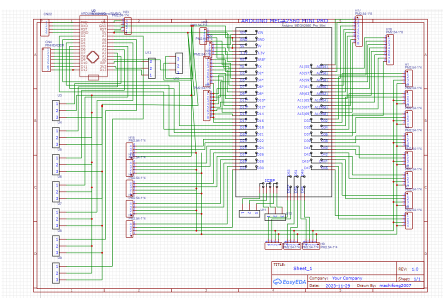


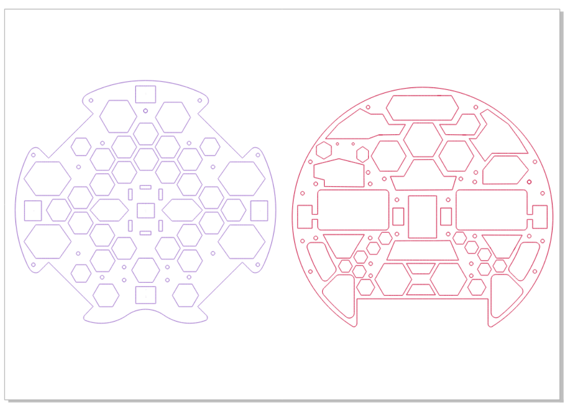


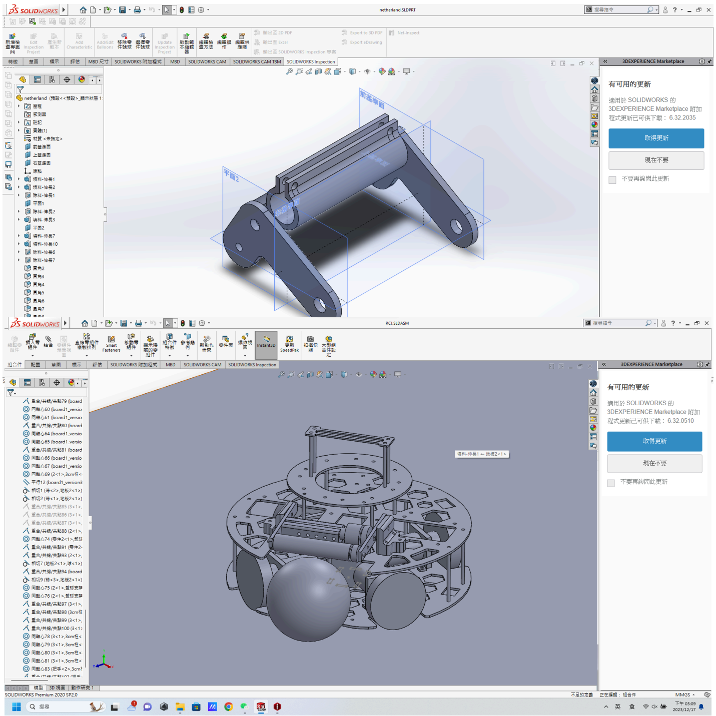
**Software - 3D design and development environment**

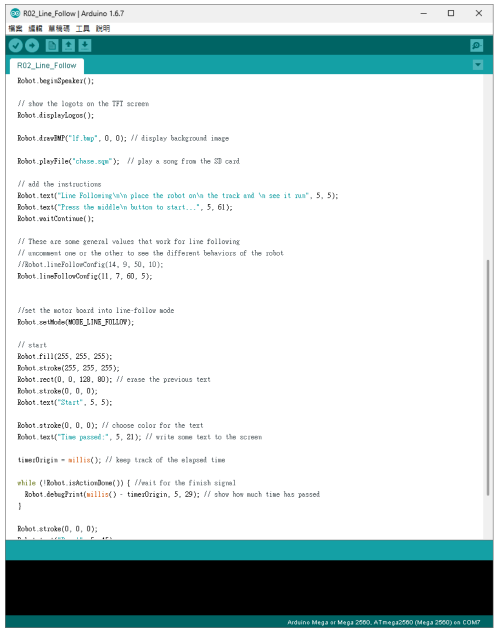
1. CorelDraw
2. SolidWorks
3. EasyEDA
4. Arduino

Our team chose several platforms for our robot design. First, CorelDRAW is used to create 2D products, such as the three layers of our robot. Then, SolidWorks is employed for 3D design, including components like the dribbling device. For circuit design, simulation, and PCB layout, we use EasyEDA, an online browser-based tool that facilitates collaborative sharing of ideas. Lastly, we discuss the use of Arduino IDE for programming due to its user-friendly interface and ease of mastery. The program itself is written in the C++ programming language





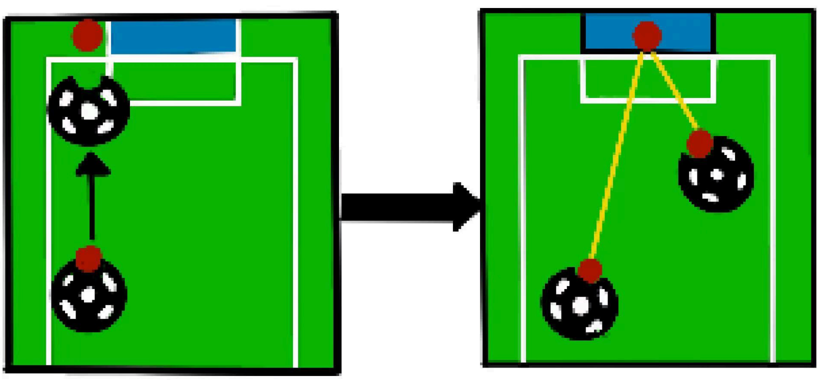




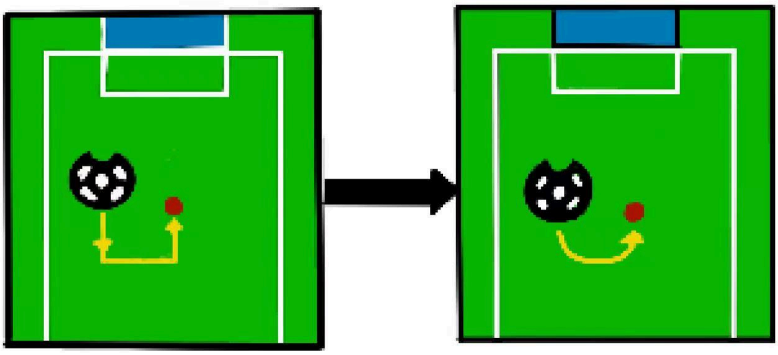

# Strategy - Programming process

Attacking robot - the striker

After extensive discussion, we divided the algorithm into two parts: handling the ball in front of the robot and behind it. For the former scenario, our initial approach was to command the robot to move forward continuously. However, we found this method to be inefficient for scoring. Consequently, we improved the code by programming the robot to keep facing the football goal. The graphs below illustrate the development and refinement of our program.

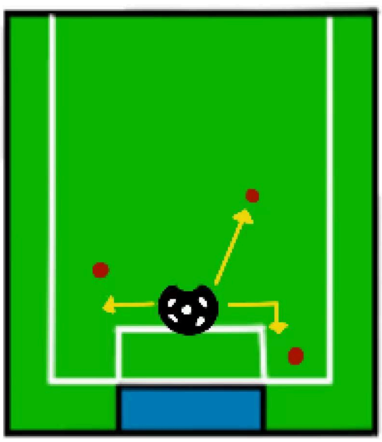


When the ball is behind the robot, our initial strategy was to have the robot move to the backside of the ball and chase it. However, this approach proved time-consuming as the robot had to move in straight lines. To optimize this, we implemented an algorithm that allows the robot to chase the ball along a curved path. The upper diagrams illustrate the process of our code improvement.



Defending robot - the goalkeeper

With two robots on the field, our team decided to designate one robot as a striker and the other as a goalkeeper. For the goalkeeper robot, the primary focus is on detecting greyscales. According to the rules, our robots must not enter the penalty area (the rectangular area in front of the goal). Therefore, our robot's movements are based on grayscale detection to ensure compliance. The graph below demonstrates the movement patterns of our defending robot during the game.

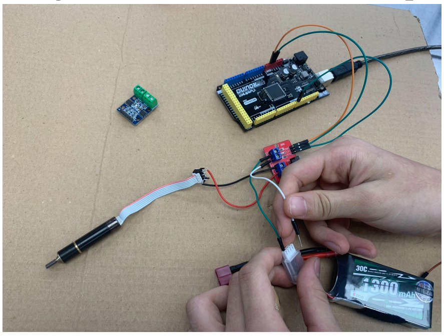


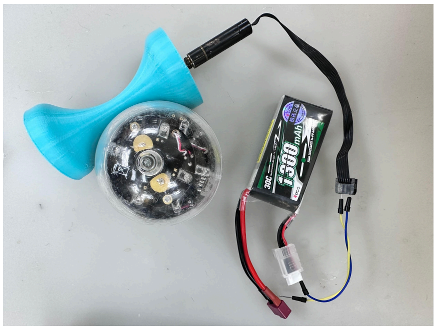
## The result

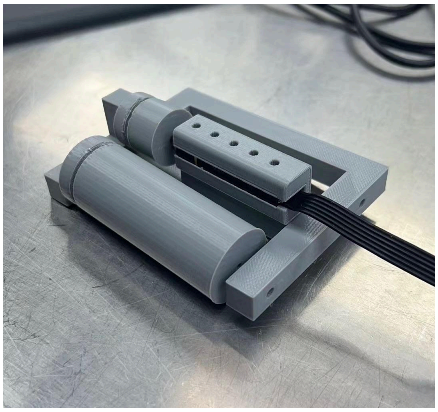
1. Hardware:

   a) Dribbling device

Since the football cannot be inside the robot for more than three centimeters, we needed a method to control the ball effectively during the game. We chose a dribbling device because it can maintain the ball in front of the robot while it moves swiftly around the field. To construct our device, we utilized 3D printing technology, an H-bridge controller, and a DC motor. We developed three versions of the device. The first and second versions featured a solid roll with a lighter motor. However, we improved the design by using a faster but heavier motor, as the weight allowance was sufficient. The final version incorporated a hollow design to better fit the ball's shape, ensuring effective control from every direction.

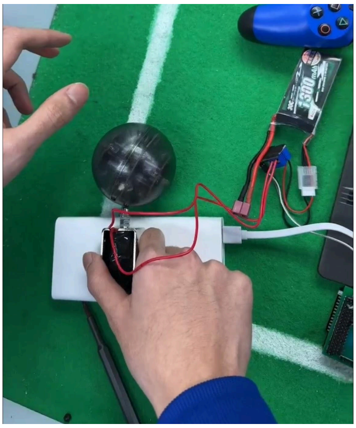


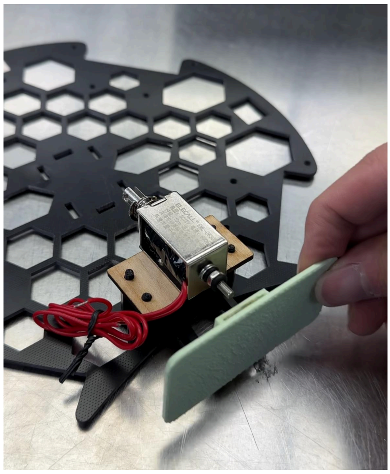




b) Batting device

A kick from the robot significantly impacts the game. An attacking robot can score more effectively, while a defending robot can clear the ball from unfavorable areas. Among the various types of motors available, we chose the electromagnet motor for its precise control and lightweight design. Additionally, we designed a flat surface to increase the contact area, optimizing the kicking mechanism.

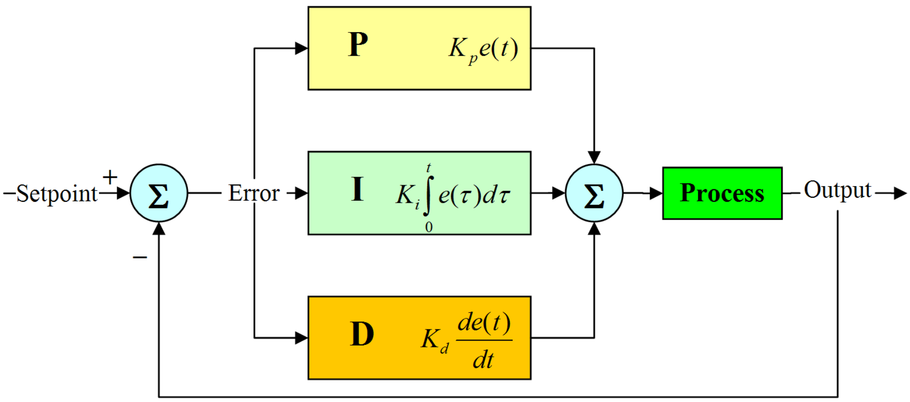




2. Programming:

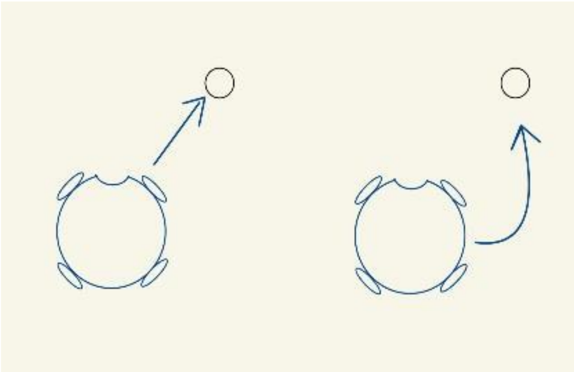
a) PID controller

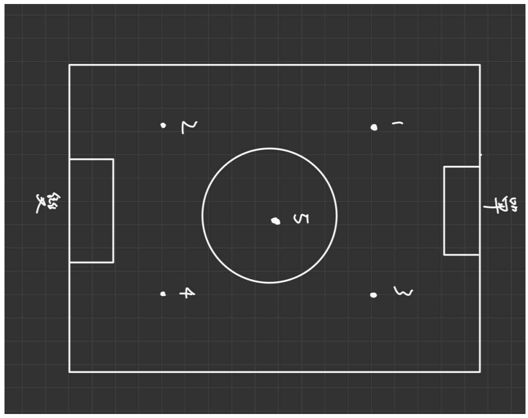
A PID controller (Proportional-Integral-Derivative controller) is a control loop mechanism widely used in industrial control systems. It continuously calculates an error value as the difference between a desired setpoint and a measured process variable. Our team applies this methodology by defining the setpoint as the direction and distance from the robot to the ball. First, the value from the IR seeker is converted into degrees and a specific value. Then, two functions calculate the speed and direction, sending the information to the PCB to control the motor's movement.




b) Ball-related movement

Instead of rushing directly to the ball, our team discussed a more effective method to control the robot and maintain possession of the ball. The robot's direction is adjusted based on the distance between the ball and the robot. As shown in the following figure, the robot moves in a curved path to position itself behind the ball first, then moves forward to control the ball using its dribbling device.

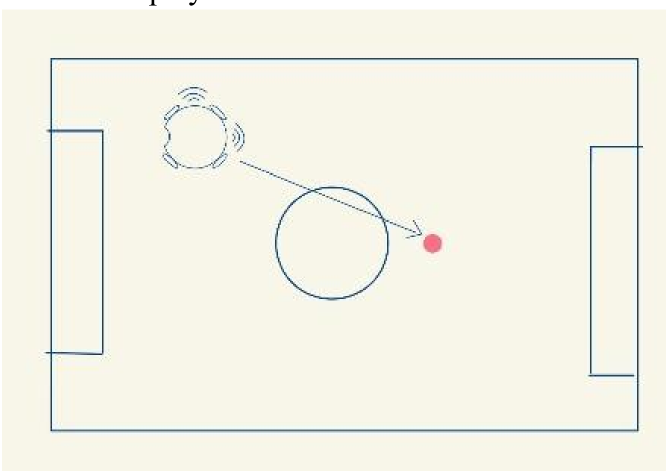




Moreover, we identified five key spots on the field, four of which are common locations where the ball is frequently positioned, and one spot in the center of the field. We trained the robot to move quickly to these spots and execute specific movements to score goals.

c) Determination of coordinates

Although we could use two if-functions to return the robot to its position when the ball is out of play, we improved the logic to shorten this duration. By calculating the robot's coordinates on a four-quadrant Cartesian plane and using inverse trigonometric functions, the angle between the robot and the target spot is determined. This approach enables the robot to move to the center of the field more efficiently, positioning it advantageously for when the ball is placed back in play.



## Conclusion and Future Work

During the creation process, we went through all stages, from engineering the robot and assembling it to programming and fine-tuning the algorithms. We learned to work with CNC and water jet cutters, design the robot using CorelDRAW and SolidWorks, cut plates with a water jet cutter, and create printed wheels using a CNC machine. Additionally, we programmed the robot, developed algorithms, and honed our critical thinking skills, even in the most challenging situations.